

Cybersecurity Threat Report (2024–2025)

Cybersecurity Internship Program

Task-1: Threat Intelligence Report

Submitted By: Vaishali A

Organization: Maincrafts Technology

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1. Introduction to Cybersecurity

What is Cybersecurity?

Cybersecurity is the practice of protecting computers, networks, servers, and data from unauthorized access, attacks, and damage.

Why is Cybersecurity Important?

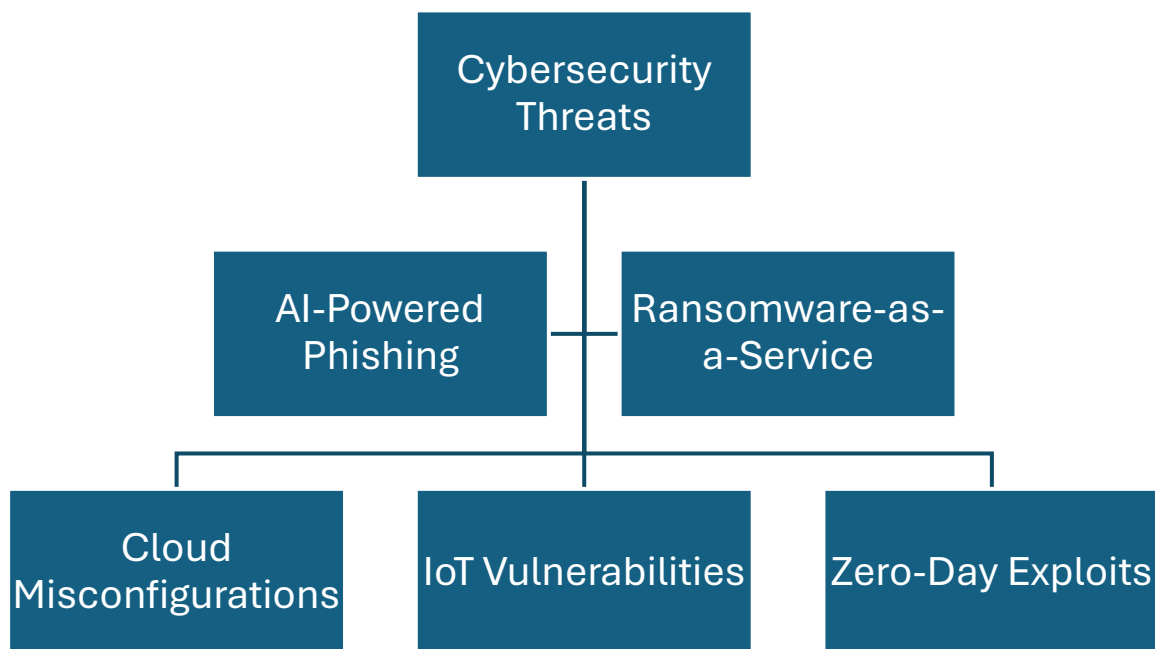
- Protects personal information
- Prevents financial losses
- Maintains business continuity
- Protects sensitive organizational data

Current Relevance

Today cybercrime is increasing rapidly because of:

- Digital transformation
- Remote work culture
- Cloud computing
- AI-driven cyber attacks

2. Major Modern Cyber Threats



Threat 1: AI-Powered Phishing Attacks

Description

AI-powered phishing uses artificial intelligence to create convincing emails, messages, and deepfake videos.

Impact

Individuals:

- Identity theft
- Financial fraud

Organizations:

- Credential theft
- Data breaches

Case Study

Arup Deepfake Scam (2024)

Employees were tricked by AI-generated video calls and millions of dollars were transferred to attackers.

Prevention

- MFA
- Employee awareness training
- Email filtering

Threat 2: Ransomware-as-a-Service (RaaS)

Description

Cybercriminals sell ransomware kits to other attackers.

Impact

Individuals:

- Data loss
- Financial loss

Organizations:

- Operational disruption
- Reputation damage

Case Study

LockBit Ransomware

One of the largest ransomware groups targeting organizations worldwide.

Prevention

- Backup data regularly
- Endpoint protection
- Patch systems

Threat 3: Cloud Security Misconfigurations

Description

Improper cloud settings expose sensitive data.

Impact

Individuals:

- Personal data exposure

Organizations:

- Data breaches
- Regulatory penalties

Case Study

Capital One Breach (2019)

Misconfigured cloud infrastructure exposed customer information.

Prevention

- Cloud audits
- Access control
- Encryption

Threat 4: IoT Vulnerabilities

Description

Internet-connected devices can be exploited by hackers.

Impact

Individuals:

- Privacy invasion

Organizations:

- Botnet attacks
- Service disruption

Case Study

Mirai Botnet

Compromised thousands of IoT devices.

Prevention

- Change default passwords
- Firmware updates
- Network segmentation

Threat 5: Zero-Day Exploits

Description

Attackers exploit software vulnerabilities before patches become available.

Impact

Individuals:

- Device compromise

Organizations:

- Data theft
- Network intrusion

Case Study

MOVEit Transfer Zero-Day Attack (2023–2024)

Attackers exploited an unknown vulnerability affecting many organizations.

Prevention

- Patch management
- Threat monitoring
- Zero Trust model

3. Impact Analysis

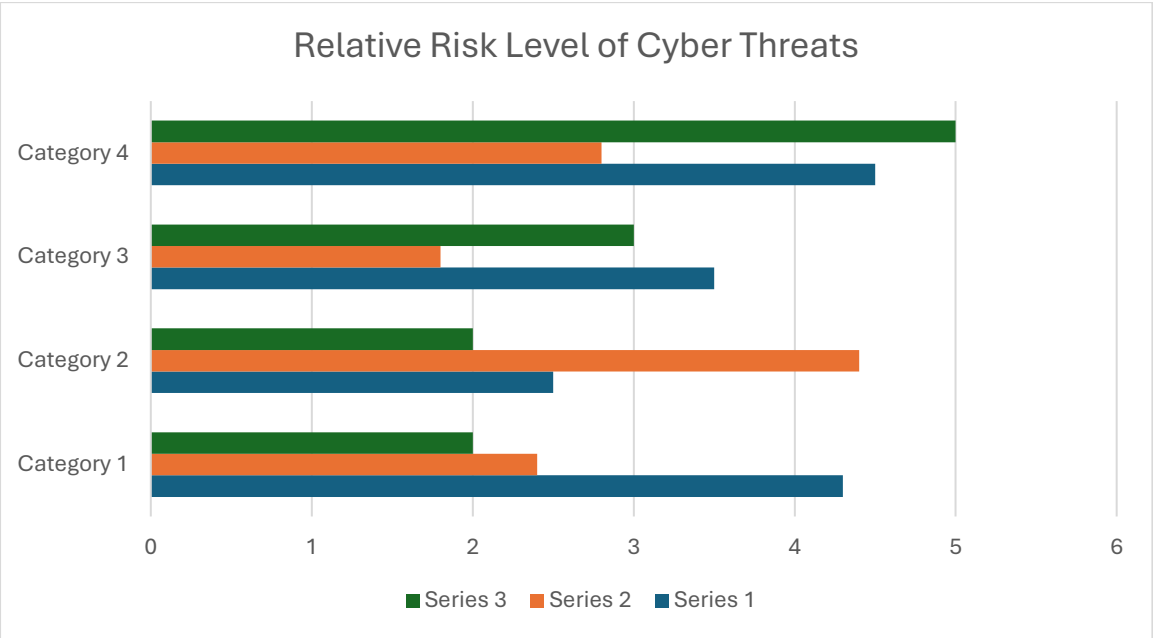
Threat	Individual Impact	Organizational Impact
AI Phishing	Financial Fraud	Data Breach
RaaS	Data Loss	Downtime
Cloud Misconfiguration	Privacy Loss	Compliance Issues
IoT Vulnerabilities	Device Hacking	Service Disruption
Zero-Day Exploits	System Compromise	Data Theft

Chart: Relative Risk Level of Threats

Data

Threat	Risk Score
AI Phishing	90
Ransomware	95
Cloud Misconfiguration	80
IoT Vulnerabilities	75
Zero-Day Exploits	98

ar Chart:



Infographic: Cyber Attack Lifecycle

Reconnaissance



Phishing / Exploitation



Unauthorized Access



Data Theft / Encryption

↓

Financial Loss

↓

Recovery & Response

4. Preventive Measures

General Security Controls

- ✓ Multi-Factor Authentication (MFA)
- ✓ Zero Trust Security Model
- ✓ Firewalls
- ✓ IDS/IPS Systems
- ✓ Security Awareness Training
- ✓ Regular Software Updates
- ✓ Strong Password Policies
- ✓ Data Encryption

5. Conclusion & Future Scope

Cybersecurity has become a critical requirement for individuals and organizations. Modern threats such as AI-powered phishing, ransomware, cloud vulnerabilities, IoT attacks, and zero-day exploits continue to evolve. Organizations must adopt proactive security measures and continuously update their defenses. Continuous learning and awareness are essential because cyber threats change rapidly with technological advancements.

6. References

- ✓ [OWASP](#)
- ✓ [CISA Cybersecurity Advisories](#)
- ✓ [IBM Security Blog](#)
- ✓ [Krebs on Security](#)
- ✓ [Microsoft Security Blog](#)